

Sean Luka Girgis

Senior Software / Data Engineer | PySpark, SQL, Python, Hadoop & Production Systems

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Professional Summary

Senior software and data engineering professional with 20+ years of enterprise experience across Python, SQL, PySpark/Spark, Hadoop/Hive concepts, Oracle/RDBMS, C/C++, REST API concepts, production troubleshooting, documentation, Agile collaboration, performance tuning, and reusable data-processing workflows. Strong background supporting enterprise systems, building reliable pipeline and reporting components, resolving production issues, optimizing SQL and application behavior, and communicating technical findings to engineering and business stakeholders. Best aligned to software/data engineering roles requiring Spark/PySpark, SQL/RDBMS, Python, legacy C/C++ depth, production support, documentation, and performance/availability improvement; Java/JEE Spring Boot/Struts/Hibernate, Elasticsearch, OpenShift/Kubernetes, and production LLM/AI solution ownership are positioned carefully as adjacent or learning areas rather than overstated production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built Python/Pandas/PySpark and SQL workflows that processed telemetry from 6,000+ infrastructure endpoints into validated dashboards, reports, forecasting datasets, and operational insights for critical banking infrastructure.
- Designed and supported ETL/ELT-style data workflows using Python, SQL, PySpark/Spark concepts, Oracle, AWS S3, Glue, Athena, and Redshift to transform raw operational data into curated reporting datasets.
- Used complex SQL across Oracle, Amazon Redshift, and reporting structures to validate transformations, troubleshoot discrepancies, optimize queries, and support production analytics.
- Troubleshoot production issues by tracing source feeds, Python/PySpark processing, SQL transformations, metric behavior, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Improved performance and availability of reporting workflows through query optimization, validation checkpoints, source-feed monitoring, runbooks, and proactive data quality checks.
- Prepared forecasting-ready datasets for Prophet and scikit-learn workflows, supporting capacity planning, trend analysis, and bottleneck prediction across enterprise infrastructure.
- Created technical documentation, metric definitions, data flow notes, support procedures, validation logic, and troubleshooting guides to support handoff and production reliability.
- Collaborated with infrastructure, application, analytics, engineering, and business teams in Agile-style delivery to clarify requirements, prioritize fixes, and communicate technical/operational issues.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Provided application performance and production visibility support for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.

- Investigated production and pre-production performance issues by correlating application monitoring data, user-impacting symptoms, and load-test behavior.
- Created dashboards and reporting views that converted monitoring signals into actionable performance, reliability, and stakeholder insights.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance data, replacing manual support/reporting workflows with repeatable processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during testing and production-readiness reviews.
- Documented performance baselines, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized C++ transaction processing and Oracle access patterns for high-volume workloads, reducing physical hardware footprint by 95% while maintaining low-latency query behavior.
- Supported schema conversion, data validation, phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using C, C++, Pro*C, Oracle, PL/SQL, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.

- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt

Skills

Languages: Python, SQL, PySpark, C, C++, PL/SQL, Perl, KornShell/ksh, Java/J2EE concepts, JavaScript awareness, C# awareness

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed an AWS data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet with validation checkpoints, SQL quality checks, and curated analytics datasets.

Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, SQL, Data Quality

ServiceCall AI RAG Demo

Built staged Python proof-of-concepts with document loading, chunking, TF-IDF retrieval, hybrid retrieval, Pydantic schemas, FastAPI service layers, Docker-first smoke tests, and deterministic API endpoints.

Technologies: Python, FastAPI, Docker, Pydantic, pytest, RAG, TF-IDF

High-Throughput Oracle RAC Migration

Led validation and production-readiness support for migration from 200+ MySQL nodes to a 6-node Oracle RAC platform, using C++/OCCI/OCI validation utilities and SQL optimization to preserve transaction correctness and performance.

Technologies: C++, OCCI, OCI, Oracle RAC, MySQL, SQL, Migration Validation

Production Data Quality and RCA Workflow

Built validation and troubleshooting workflows for production reporting datasets, including row-count checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and documented recovery steps.

Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA